ANYA ARORA

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LeetCode ◇ LinkedIn ◇ GitHub

EDUCATION

B.E. – Data Science & Artificial Intelligence *2024 – 2028*
Thapar Institute of Engineering & Technology, Patiala | GPA: 9.43/10

Class XII (CBSE) *2012 – 2024*
Amity International School, Sector 46, Gurgaon | 93%

PROJECTS

DashQL - Agentic Analytics | *Python, LangGraph, OpenAI, ChromaDB, Streamlit* Live | GitHub | *Jul 2026*

- Built an agentic LLM workflow with a 12 node architechture using **LangGraph** and **OpenAI API** transforming natural-language requests into SQL-driven dashboards and automated insights at **~$0.005**/dashboard; added **end-to-end observability** across the agentic workflow.
- Implemented **automated SQL validation and governance** using AST-based checks (sqlglot), enforcing read-only, PII-safe execution and **prompt-injection defenses**; built a RAG-based caching system with ChromaDB and GitHub-backed dashboard storage, incorporating **confidence-aware retrieval and cache invalidation** to reduce latency from 11.4s to 5.3s on cache hits.
- Built a 100-query **evaluation and testing benchmark** for LLM-generated SQL, achieving 91.67% database accuracy, 100% governance pass rate, and 4.59/5 LLM-as-a-Judge correctness;

SplitCore - Expense Sharing App | *Flask, MySQL, HTML, CSS, JavaScript* Live | GitHub | *Nov 2025*

- Designed and developed a **full-stack** back-end service (Flask, JavaScript) with RESTful API endpoints, **bcrypt authentication** , expense-sharing workflows, and session-based access control for groups of up to 20 users.
- Implemented **Pydantic-based API input validation**, enforcing data constraints and handling invalid inputs with HTTP errors.
- Architected a **normalised relational schema (3NF)** across 5 tables with soft-delete and settlement tracking, and developed a **greedy debt-simplification algorithm** that reduced redundant payments by up to **70%**.

Quora Duplicate Question Detection | *Python, NLP, LightGBM, BERT* GitHub | *Feb 2026*

- Built an end-to-end **NLP data processing pipeline** over **400K+ records**, using **regex based preprocessing** and feature engineering with **TF-IDF/Word2Vec**; trained a **LightGBM** classifier achieving **82% accuracy**.
- Fine-tuned **BERT** on domain-specific question pairs, improving performance to **91% accuracy** and **0.88 F1-score**.

TECHNICAL SKILLS

Languages	Python, C/C++, SQL, HTML, CSS, JavaScript
Libraries	NumPy, Pandas, Matplotlib, Seaborn, TensorFlow, OpenCV, Pydantic
Frameworks	Flask, FastAPI, LangChain, LangGraph
Database	MySQL, SQLite, SQLAlchemy
AI/ML	LLMs, Generative AI, Agentic AI, RAG, NLP
Tools	Git, GitHub, VS Code, Jupyter, Google Colab
Coursework	Data Structures & Algorithms, DBMS, OS, OOPS, Software Development

ACHIEVEMENTS

- Cleared 3 competitive rounds to reach National Finals in the JP Morgan Code For Good Hackathon among more than 50,000 participants.
- Awarded a merit scholarship for 2 consecutive years, reflecting a well-rounded academic background.
- Collaborated in cross-functional teams across 5+ hackathons to design and prototype solutions under tight deadlines, mirroring the cross-team, deadline-driven work central to this program.
- Led the college badminton team to **3 gold medals** across inter-college tournaments, demonstrating team leadership and competitive performance.